

Hardware Processing Engines on FPGA

Evaluating open-source RISC-V platforms for Edge AI accelerators

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Motivation: Edge AI is power- and area-constrained

The field of AI has had several breakthroughs in the past decade leading to widespread adoption in almost every field. Many such applications are constrained by power($\approx$ milliwatts) and area($\approx$ square millimeters) of the inference unit.

Examples:

- ▶ **Rail fault detection** — on-track, battery-powered units.
- ▶ **Breast cancer screening** — handheld imaging devices.
- ▶ **Driver assistance (ADAS)** — latency bound.

Each workload is matrix-heavy and real-time, but runs on a very small budget.

Four constraints define the design space

1. **Power.** A few hundred milliwatts in total
2. **Area.**
3. **Extensibility.** Allow for custom workloads and optimization of these workloads.
4. **Toolchain.** A C compiler that can emit instructions for the hardware.

Edge-AI hardware is fragmented across every axis

A unified platform would have to span the following combinatorial space:

CPU cores	RI5CY / CV32E40P, CV32E40PX, CVA6, Snitch, Ibex, Rocket / BOOM, OpenC910, Arm Cortex-M, Tensilica Xtensa
Accelerators	Quadrilatero, Gemini, RedMule, NE16 / NEUREKA, ITA, VTA, Coral NPU, Hailo-8, Ethos-U, custom HWPEs
Accel. interfaces	CV-X-IF, HWPE-Stream, RoCC, SSR / FREP, AXI / AXI-Stream, OBI, TileLink, vendor-private
FPGA targets	Nexys-A7 (Artix-7), Genesys2 (Kintex-7), ZCU102 (UltraScale+), VCU118, PYNQ-Z2, Arty, Intel DE-series, Lattice ECP5
Compilers	GCC (riscv-gnu), LLVM / Clang, xheep_matrix_spec LLVM, vendor SDKs, TFLite Micro, TVM, Exo, Deeploy, MLIR
Simulators / EDA	Verilator, QuestaSim, VCS, Vivado / Vitis, GVSoC, Spike, QEMU, Renode, FuseSoC, Bender

Cores $\times$ *Accels* $\times$ *Interfaces* $\times$ *FPGAs* $\times$ *Compilers* $\times$ *Simulators* — no single platform spans it.

Part I.

The platform survey

Platform 1: Coral NPU

Evaluated as a candidate for rapid edge-AI prototyping (Feb 2026).

Issues.

- ▶ No FPGA flow.
- ▶ No RISC-V Matrix core.

Verdict. Abandoned. Pivoted to the open-source PULP ecosystem.

Platform 2: PULPissimo — toolchain is too old

Hardware is sound. The software around it has gone end-of-life.

Issues.

- ▶ SDK requires gcc-5 (only Ubuntu 18); fails on modern hosts.
- ▶ RTL simulation requires QuestaSim/Xcelium(not open-source); ModelSim trial fails on common_cells(not officially supported).
- ▶ Supported FPGAs not available in RISE Lab.

```
$ make build
make[2]: Leaving directory '/home/Myuser/riscv/pulp/rtl/tb/
remote_bitbang'
** Warning: ../ips/common_cells/src/fifo_v3.sv(13): (vlog-2186)
SystemVerilog testbench feature
(randomization, coverage or assertion) detected in the design.
These features are only supported in Questasim.
```

Verdict. Abandoned — reproducibility unacceptable for developing applications.

Platform 3: PULP-derived alternatives

Platform	Blocker	Status
Chimera	Documentation thin; no FPGA flow.	rejected
Siracusa	No public repository.	rejected
Mempool + ITA	FPGA infeasible at full scale.	rejected
Snitch Cluster	No FPGA support out of the box.	rejected
GAP9	Proprietary; not open-source.	rejected
Cheshire	FPGA flow exists; no Deeploy backend.	set aside

Common limitation: no ready-made FPGA synthesis flow combined with an accelerator-integration pipeline.

Platform 4: X-HEEP + Quadrilatero — adopted

X-HEEP is a configurable RISC-V microcontroller (by EPFL-ESL) derived from the PULP ecosystem.

Quadrilatero is a systolic-array co-processor. Communicate over CV-X-IF.

- ▶ **FPGA-first.** First-class support for Nexys-A7 and Genesys2.
- ▶ **Active toolchain.** Docker image, Makefile flow, Verilator-support.
- ▶ **CV-X-IF interface.** Co-processors attach without modifying MCU RTL. Incredible eXtensibility.
- ▶ **Quadrilatero.** Open systolic array implementing the custom `xtheadmatrix` ISA extension.

Confirmed as the project target after consultation with Dr. Karthik.

Part II.

Bringing up the system

System Architecture

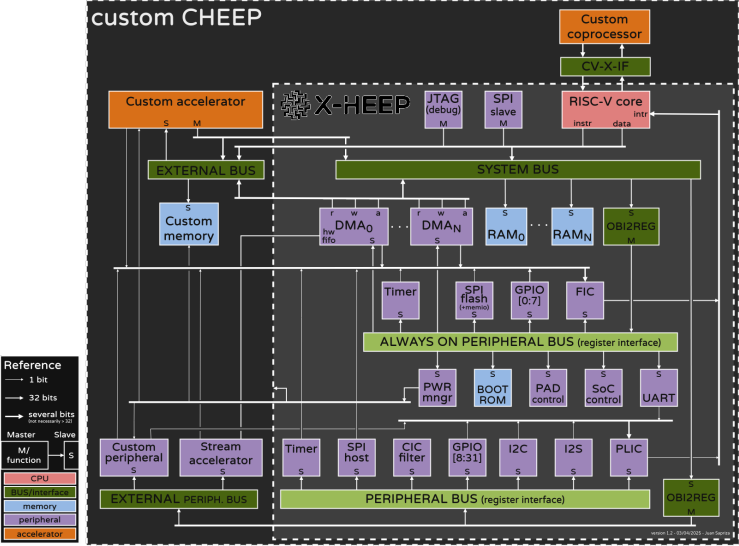


Figure: X-HEEP Microcontroller

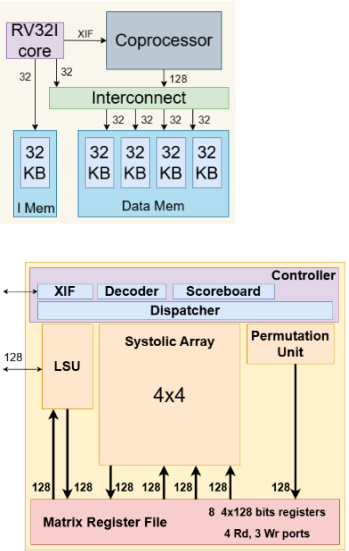


Figure: Coprocessor 11 / 24

System Architecture

- ▶ **CPU:** CV32E40PX, a 32-bit 4-stage inorder processor developed by PULP.
- ▶ **Coprocessor:** Quadrilatero, A 4x4 systolic array co-processor.
- ▶ **Interface:** CV-X-IF, a RISC-V extension interface to facilitate communication between a CPU and a coprocessor.

Architecture

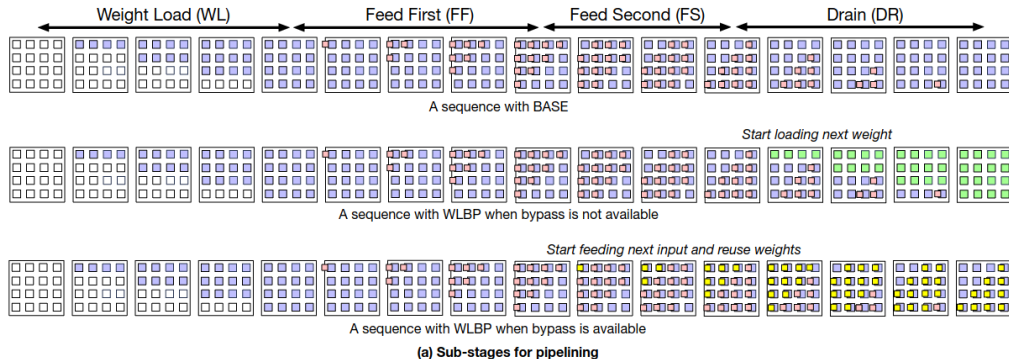


Figure: RASA pipeline stages

The systolic array is a Weight Static (WS) systolic array inspired from Register-Aware Systolic Array Matrix Engine (RASA).

Layer 1: FPGA bitstream

Custom Dockerfile on top of x-heep-toolchain, with Vivado *mounted* as an external volume rather than bundled.

Issue — Vivado version. The FPGA scripts use `-freq_hz` in `create_bd_port`, introduced after 2018.3.

```
Unknown option '-freq_hz', please type 'create_bd_port -help' for usage info.  
while executing "source xilinx_generate_clk_wizard.tcl"
```

Resolution. Mount Vivado 2022.1+; then:

```
$ make mcu-gen  
$ make vivado-fpga FPGA_BOARD=nexys-a7-200t
```

produces a complete bitstream.

Layers 2 & 3: Simulator and compiler

Verilator (COMBDLY). Verilator 5.x escalates a non-blocking-in-comb warning to an error; found in CV32E40PX clock-gate model. Fix:

```
/* verilator lint_off COMBDLY */  
if (clk_i == 1'b0) clk_en <= en_i | scan_cg_en_i;  
/* verilator lint_on COMBDLY */
```

LLVM (xtheadmatrix). The extension is not in upstream LLVM; build the patched compiler from xheap_matrix_spec and install into a user-owned mounted directory (and not /usr due to ownership issues):

```
$ make app PROJECT=example_matmul_quadilatero \  
    ARCH=rv32imc_zicsr_xtheadmatrix0p1 \  
    COMPILER=clang CLANG_LINKER_USE_LD=1
```

Part III.

The Quadrilatero bug

Symptom: result-valid never asserted

Compiled `example_matmul_quadilatero`, simulated under Verilator, inspected waveforms in GTKWave.

Observed:

1. CPU dispatches every `xtheadmatrix` instruction over CV-X-IF. ✓
2. Quadilatero accepts each instruction (`accept = 1`). ✓
3. *Result-valid is never asserted.* ✗

This rules out interface-level errors. The issue lies inside Quadilatero.

Root cause: valid-ready deadlock; both sides waited

Before — deadlock.

- ▶ RF sequencer: waited for `ready` before asserting `valid`.
- ▶ Systolic array: waited for `valid` before asserting `ready`.

Underlying cause: the `W`, `A`, `D` registers load from memory with non-zero latency; the original code held `valid` low until all three landed.

Fix.

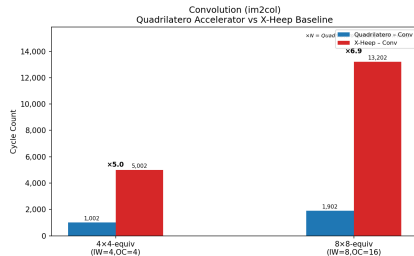
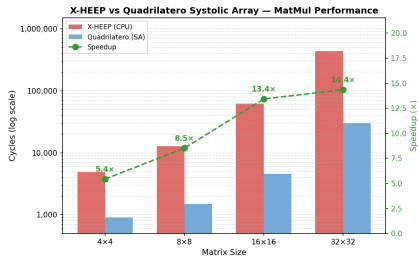
- ▶ RF sequencer asserts `valid` per individual register, as each loads.
- ▶ Systolic array asserts `ready` only once *all three* (`W`, `A`, `D`) have arrived.

ready-valid handshake semantics fixed. Correct results for matrix inputs up to 16×16 ; 32×32 works after relinking at `0x10000`; 64×64 still constrained by the SRAM bank configuration.

Part IV.

Conclusion and Future work

Results



Results don't match with Quadrilatero paper (~5x). Maybe because I turned off WLS (Weight Load Select) optimization due to correctness issues.

Delivered.

- ▶ Reproducible Docker build environment.
- ▶ Working Verilator simulation; GEMM and convolution verified.
- ▶ RTL fix for the Quadrilatero handshake deadlock.
- ▶ Generated bitstream for the Nexys-A7-200t.

Next.

- ▶ FPGA validation against the simulation.
- ▶ Deploy → `xtheadmatrix` integration.

Lessons Learned

- ▶ Software is the **bottleneck**.
- ▶ High barrier of entry for engineers trying to solve **real-world** problems.
- ▶ Dependency hell is **real**.
- ▶ Docker containers **rock**.
- ▶ Research in Edge AI should be **easy** to reproduce.

Source code

All work done as part of this project is available at:
<https://github.com/kuku929/QuadxHeep/tree/main>

Platform	Period	Key finding	Status
Google Coral NPU	Feb 2026	Closed ASIC; no RTL, no FPGA flow, no custom ISA. Only TFLite-to-Coral path.	Abandoned
PULPissimo	Mar 9–30, 2026	Hardware sound, but SDK end-of-life (gcc-5 only) and RTL simulation requires QuestaSim.	Abandoned
Chimera	Mar 13, 2026	Multi-HWPE template; documentation thin, no FPGA flow.	Rejected
Snitch Cluster	Mar 2026	Verilator-clean, Docker available, but no FPGA support out of the box.	Rejected
Cheshire	Mar 30, 2026	Linux-class RISC-V with FPGA flow, but no Deeploy backend.	Set aside
X-HEEP + Quadrilatero	Mar 31 – May 2026	Active Docker toolchain, CV-X-IF interface; bitstream + simulation working; RTL handshake bug fixed.	Adopted

Thank you. Questions?